Personalized Diet and Daily Routine Recommendation System

A Machine Learning & Conversational AI-Based Mobile Wellness Platform

UQ25CA741A - Capstone Project - Phase 1 (80%+ Advanced Milestone)

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Presentation Overview

1. Abstract
2. Introduction and problem scenario
3. Literature survey
4. Proposed solution
5. Functional and non-functional requirements
6. System architecture
7. ML methodology and process flow
8. Database design
9. Mobile implementation demonstration
10. Security and offline support
11. Tools and technologies
12. Applications, future scope and conclusion
13. References

Abstract

Maintaining a healthy diet and daily routine is difficult because users have different bodies, schedules, goals, food preferences and restrictions. Existing applications often provide generic plans or separate tracking features.

This project develops a Flutter-based Android mobile application that creates a personalized wellness plan from health profile information: age, gender, height, weight, BMI, activity level, goal, food preference and allergies.

It calculates BMI, BMR, calorie and hydration targets, and provides meal plans, recipes, substitutions, grocery planning, water, exercise, sleep and meal adherence tracking, wellness scoring and offline access.

CURRENT PROTOTYPE

Flutter Android app with local data, local rules and device persistence

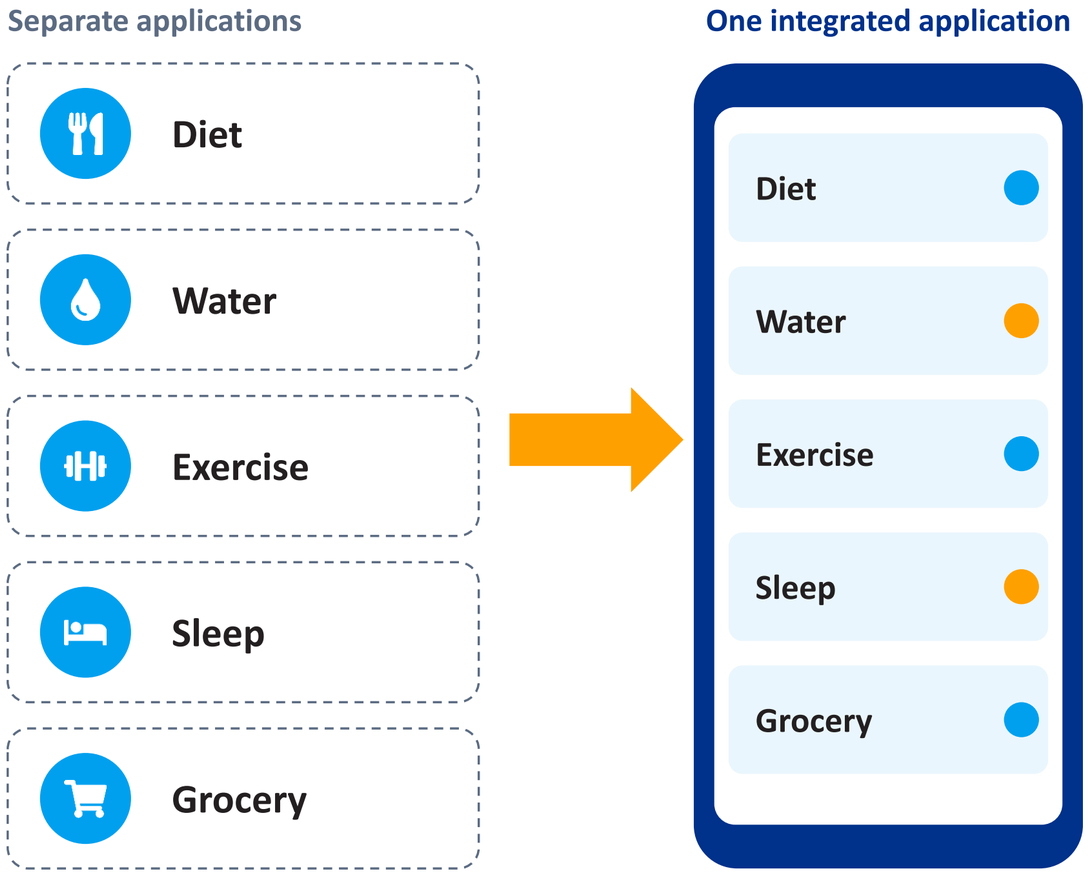
PLANNED NEXT STAGE

Flask REST API, MySQL database and Scikit-learn recommendation engine

The application provides general wellness guidance and is not a substitute for medical or dietetic advice.

Introduction and Problem Scenario

1. People find it difficult to plan meals according to BMI, calorie requirements and health goals.
2. Generic diet charts do not adapt to weight changes, activity levels, dislikes or allergies.
3. Users may not know suitable substitutes when ingredients are unavailable.
4. Grocery planning is usually a separate manual activity.
5. Water, exercise, sleep and meal tracking are spread across different applications.
6. Users need one mobile application that combines planning, tracking and adaptive guidance.



Proposed Solution, Purpose and Scope

• Dynamic Macronutrient Balancing: Visual real-time distribution across Protein (25%), Carbohydrates (50%), and Fats (25%).

Purpose

- Improve personalization compared with static diet charts.
- Encourage consistent daily habits.
- Reduce the effort required to plan meals and groceries.
- Provide useful substitutes for unavailable or disliked foods.
- Keep previous plans and logs available offline.

Scope

- Android mobile application using Flutter.
- Local-first prototype for profile, meals, recipes, grocery and tracking.
- Planned Flask API, MySQL database and ML engine.
- General wellness use for students, professionals and fitness-conscious users.

Domain and Target Users

Domain areas

- Digital health and mHealth
- Nutrition informatics
- Recommender systems
- Supervised machine learning
- Mobile computing
- Habit and wellness tracking

Target users

- Students
- Working professionals
- Fitness-conscious users
- Gym members and trainers
- Sports departments in educational institutions
- Workplace wellness programmes

Medical professionals and dietitians remain important for condition-specific advice.

Literature Survey Background

Mobile health applications for lifestyle monitoring

Food and recipe recommender systems

Nutrient-based content recommendation

Machine learning for calorie and dietary prediction

Behaviour change and habit tracking applications

Offline-first mobile health applications

Requirement: include at least ten verified papers from IEEE Xplore, Scopus-indexed journals or recognised academic conferences, preferably from the last five years. Use only details verified through the university library or IEEE/Scopus access.

Literature Survey Summary (1/5)

No.	Paper title	Author(s)	Year / venue	Main method	Main finding	Relevance to project
1	**The Use of Social Features in Mobile Health Interventions to Promote Physical Activity: A Systematic Review**	H. L. Tong and L. Laranjo	2018, npj Digital Medicine	Systematic review and meta-analysis	Social support and comparison can improve engagement, but user preferences differ.	Supports reminders, progress feedback and future social or motivational features.
2	**Mobile Phone Apps to Promote Weight Loss and Increase Physical Activity: A Systematic Review and Meta-Analysis**	G. Flores Mateo, E. Granado-Font, C. Ferre-Grau and X. Montana-Carreras	2015, Journal of Medical Internet Research	Systematic review and meta-analysis	Mobile applications can support weight management and physical activity, although study quality varies.	Supports combining diet, exercise, water and progress tracking in one app.

Literature Survey Summary (2/5)

No.	Paper title	Author(s)	Year / venue	Main method	Main finding	Relevance to project
3	**A Systematic Review of Electronic and Mobile Health Physical Activity Interventions for African American and Hispanic Women**	R. P. Joseph, K. E. Royse and T. J. Benitez	2019, Journal of Physical Activity and Health	Systematic review of eHealth and mHealth interventions	Personalisation and culturally appropriate content are important for engagement.	Supports user-specific goals, food preferences and regional food support.
4	**The Effectiveness of e- and mHealth Interventions to Promote Physical Activity and Healthy Diets in Developing Countries: A Systematic Review**	A. M. Muller, S. Alley, S. Schoeppe and C. Vandelanotte	2016, International Journal of Behavioral Nutrition and Physical Activity	Systematic review	Mobile interventions can improve lifestyle behaviours, but implementation and access affect outcomes.	Supports an affordable Android application with offline access.

Literature Survey Summary (3/5)

No.	Paper title	Author(s)	Year / venue	Main method	Main finding	Relevance to project
5	**Affective Video Recommender Systems: A Survey**	D. Wang and X. Zhao	2022, Frontiers in Neuroscience	Survey of affective recommendation methods	User context and preference signals can be used to improve recommendation quality.	Supports future context-aware routine and meal recommendations.
6	**Recommender Systems and Their Ethical Challenges**	S. Milano, M. Taddeo and L. Floridi	2020, AI & Society	Ethical analysis of recommender systems	Recommenders must consider autonomy, bias, transparency, privacy and manipulation risks.	Supports explainable recommendations, privacy controls and the medical-advice limitation.

Literature Survey Summary (4/5)

No.	Paper title	Author(s)	Year / venue	Main method	Main finding	Relevance to project
7	**What We Talk About When We Talk About Trust: Theory of Trust for AI in Healthcare**	F. Gille, A. Jobin and M. Ienca	2020, Intelligence-Based Medicine	Conceptual analysis of trust in healthcare AI	Trust depends on transparency, accountability, competence and user expectations.	Supports showing calculation logic and clearly stating model limitations.
8	**Explainable Recommendation: A Survey and New Perspectives**	Y. Zhang and X. Chen	2020, Foundations and Trends in Information Retrieval	Survey and taxonomy of explainable recommendation	Explanations can improve transparency and user understanding beyond accuracy alone.	Supports explaining calorie targets, meal choices and substitutions.

Literature Survey Summary (5/5)

No.	Paper title	Author(s)	Year / venue	Main method	Main finding	Relevance to project
9	**Latest Trends of Security and Privacy in Recommender Systems: A Comprehensive Review and Future Perspectives**	A. Himeur et al.	2022, Computers & Security	Security and privacy survey	Recommender systems require privacy-aware data collection, access control and secure processing.	Supports JWT, password hashing, HTTPS and user-specific data access.
10	**Ethical and Legal Analysis of Machine Learning Based Systems: A Scenario Analysis of a Food Recommender System**	O. Levina and S. Mattern	2023, Recommender Systems: Legal and Ethical Issues, Springer	Scenario-based ethical and legal analysis	Food recommenders raise issues involving privacy, fairness, accountability and social impact.	Supports allergen filtering, safety checks, privacy design and professional-advice warnings.

Comparison with Existing Applications

Feature	Generic diet chart	Calorie counter	Recipe application	Fitness tracker	Proposed application
BMI and BMR calculation	✗	Partial	✗	Partial	✓ Both calculated
Personalized calorie target	✗	✓	✗	Partial	✓ BMR x activity + goal
Adaptive meal plan	✗	✗	✗	✗	✓ Updates with profile and progress
Recipe recommendation	✗	✗	✓	✗	✓ Recipe for every meal
Food substitution	✗	✗	Partial	✗	✓ Nutrition-based alternatives
Grocery list generation	✗	✗	Partial	✗	✓ From weekly meals
Water tracking	✗	Partial	✗	✓	✓ Quick-add logging
Exercise tracking	✗	Partial	✗	✓	✓ Duration logging
Sleep tracking	✗	✗	✗	✓	✓ Duration logging
Wellness score	✗	✗	✗	Partial	✓ Daily score from 0 to 100
Reminders	✗	Partial	✗	✓	✓ Reminder preferences
Offline access	✓	Partial	Partial	Partial	✓ Plans and logs stored locally
Progress analytics	✗	✓	✗	✓	✓ Charts of daily progress
One integrated workflow	✗	✗	✗	✗	✓ Planning + tracking in one flow

Typical capabilities of each category; individual products vary. The proposed application combines these capabilities into one flow.

Functional Requirements

1. User registration, multi-role auth (Member/Admin) & guest login
2. Health profile creation, biometrics & target goal weight
3. BMI, BMR (Mifflin-St Jeor) & calorie deficit calculations
4. Real-time macronutrient balance bars (P:25%, C:50%, F:25%)
5. Adaptive 5-meal daily nutrition plan generation
6. Recipe details, ingredients & cooking preparation display
7. Content-based nutrient smart substitutions
8. Smart grocery checklist with category filtering & sharing
9. Circadian daily routine timeline (7 scheduled milestones)
10. Bidirectional tracking steppers (Water +/-250ml, Exercise +/-10m)
11. Sleep duration steppers (+/-1hr) & restorative sleep tracking
12. Main meal adherence compliance cards (+15 pts gamification)
13. Daily Wellness Score calculation (0–100 weighted index)
14. Conversational AI Coach (Gemini 1.5 Flash API & local engine)
15. 7-Day habit consistency analytics bar chart
16. Admin Console: Member health audit & recipe catalog manager
17. Offline-first PWA caching (Service Worker v9 & LocalStorage)
18. Cross-platform mobile store packaging (WebAPK & Android AAB)

Functional Requirement Details

Requirement	Input	Processing	Output
Health profile	Age, height, weight, activity and goal	Calculate BMI, BMR and calorie target	**Health summary**
Meal plan	Profile and target calories	Split calories across five meals	**Daily meal plan**
Substitution	Selected food	Compare nutrition and category	**Alternative food suggestions**
Grocery list	Weekly meals and ingredients	Aggregate quantities	**Shopping checklist**
Wellness score	Meal, calorie, water, exercise and sleep logs	Weighted calculation	**Score from 0 to 100**
Remaining requirements follow the same input, processing and output pattern.			
AI Wellness Coach	Natural language health query	Google Gemini 1.5 Flash / Local Intel Engine	Personalized diet advice & swaps
Daily Routine	7 circadian milestone toggles	Real-time compliance state tracker	Habit adherence ratio & streak

Non-Functional Requirements

Usability
Simple mobile interface with clear navigation.

Performance
Common local actions should respond immediately.

Availability
Saved plans and logs should be available offline.

Security
Passwords must be hashed and API calls authenticated.

Privacy
Only necessary health data should be collected and returned.

Maintainability
Separate UI, model, persistence and recommendation layers.

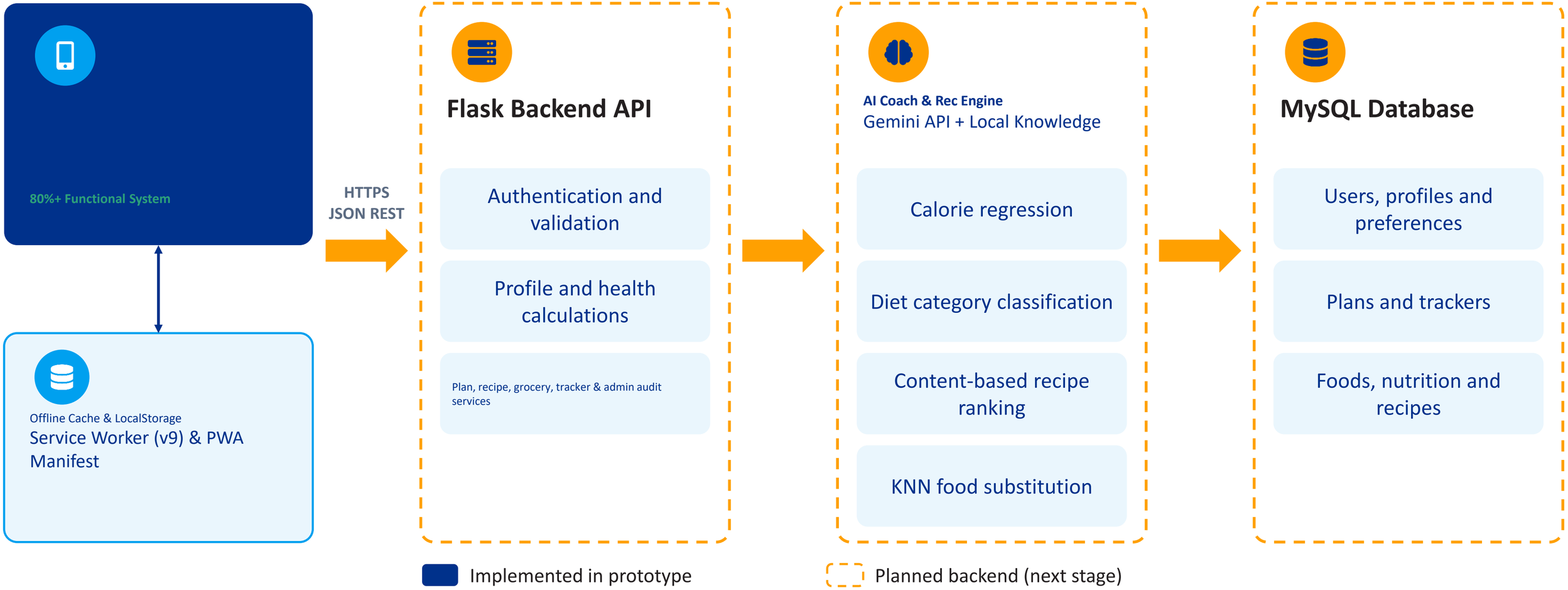
Scalability
Backend should support more users, foods and recipes.

Reliability
Invalid values must be rejected and allergen violations prevented.

Portability
Flutter allows future iOS support.

Explainability
Calorie and wellness results should be understandable to users.

System Architecture



Mobile Application Architecture

Presentation Layer	Dashboard, Plan, Track (Circadian Timeline), Grocery, Profile, Admin Console, AI Coach Modal	**Prototype**
Application Layer	Wellness scoring, macro balance algorithms, bidirectional steppers, AI chat handler, routine state engine	**Prototype**
Domain Layer	HealthProfile, MealCatalog, GroceryItem, DailyRoutineSchedule, AdminDirectory, ChatSession	**Prototype**
Persistence Layer	SharedPreferences now; SQLite sync queue in the backend phase	**Now + planned**
Remote Layer	Flask REST API planned for server synchronization	**Planned**

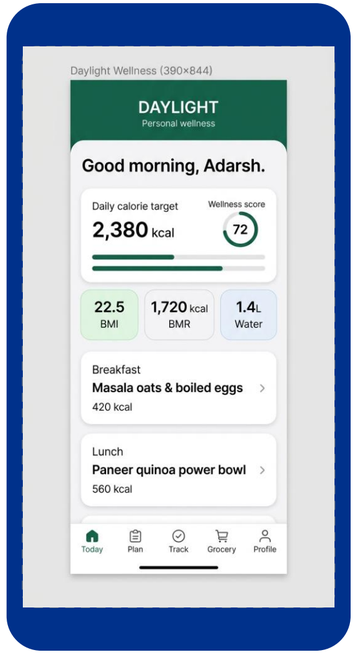


Replaceable engine

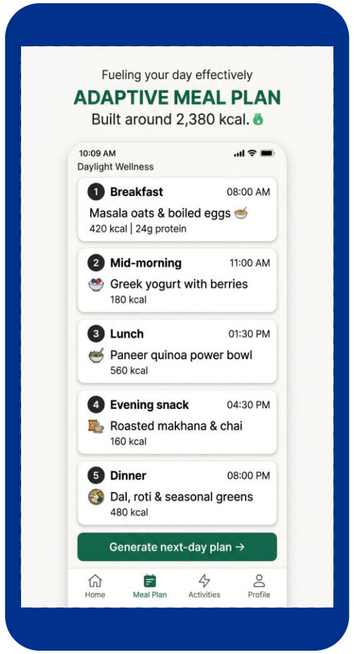
This separation makes the local recommendation engine replaceable by a server ML service later.

UI and storage stay unchanged; only the engine behind the domain layer is swapped.

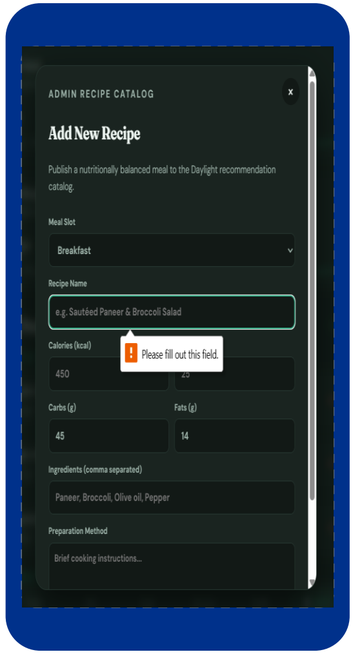
Implementation Demonstration



Dashboard



Meal plan



Daylight AI Coach

Live mobile screenshots from Daylight Wellness PWA running standalone with offline service worker v9.

Working flows to demonstrate

1. Multi-role Sign-in, Register & Guest Access
2. Dashboard: Calories, Macros, BMI & Wellness Score
3. Editable Health Profile & Target Goal Weight
4. Adaptive 5-Meal Plan with Smart Swaps
5. Recipe Details, Ingredients & Preparation Method
6. Smart Food Nutrient Substitutions
7. Categorized Grocery Checklist & Share List
8. Bidirectional Water Intake Steppers (+/- 250ml)
9. Bidirectional Exercise & Sleep Duration Steppers
10. Main Meal Adherence Cards (+15 pts gamification)
11. Circadian Routine Timeline (7 Milestones)
12. Conversational AI Coach (Gemini API & Offline)
13. Admin Console: User Audits & Recipe Manager

Health Calculation Methodology

1 BMI

$$BMI = weight\ (kg) / [height\ (m)]^2$$

2 Mifflin-St Jeor BMR

$$Male\ BMR = 10W + 6.25H - 5A + 5$$
$$Female\ BMR = 10W + 6.25H - 5A - 161$$

3 Daily calorie target

$$Daily\ target = BMR \times activity\ factor + goal\ adjustment$$

4 Hydration target

$$Water\ target = body\ weight\ (kg) \times 35\ ml$$

W = weight (kg) H = height (cm) A = age (years)
Activity factor depends on activity level. Goal adjustment changes for weight loss, maintenance or muscle gain.

ML and Recommendation Process Flow

1. Collect user profile
2. Validate values and restrictions
3. Calculate BMI, BMR and base calorie target
4. Create feature vector
5. Predict calorie target and diet category
6. Split calories across five meals
7. Filter foods by preference and allergies
8. Rank foods and recipes by nutrient similarity
9. Generate substitutions
10. Create routine, hydration target and grocery list
11. Store plan and track user feedback
12. Regenerate plan when profile or progress changes

ML step in planned backend (local rules in prototype)

Current prototype

Uses local rules and sample meal data.

Planned backend

Loads trained Scikit-learn models.

Model updates

Models are retrained offline rather than learning directly from every request.

Pseudocode

```
INPUT profile

bmi = weight / height_metre^2
bmr = calculate_mifflin_st_jeor(profile)
calorie_target = bmr * activity_factor(profile.activity)
calorie_target = calorie_target + goal_adjustment(profile.goal)
water_target = weight * 35

meal_targets = split_into_five_meals(calorie_target)

FOR each meal_target:
    candidates = foods_matching_preference(profile.preference)
    candidates = remove_allergens(candidates, profile.allergies)
    selected_food = rank_by_nutrient_similarity(candidates, meal_target)
    attach_recipe(selected_food)

score = calculate_wellness_score(meal_logs, water, exercise, sleep)
grocery_list = aggregate_recipe_ingredients(meal_plan)

OUTPUT plan, recipes, substitutions, grocery_list, score
```

1 Compute targets

BMI, BMR, calorie target and water target from the profile.

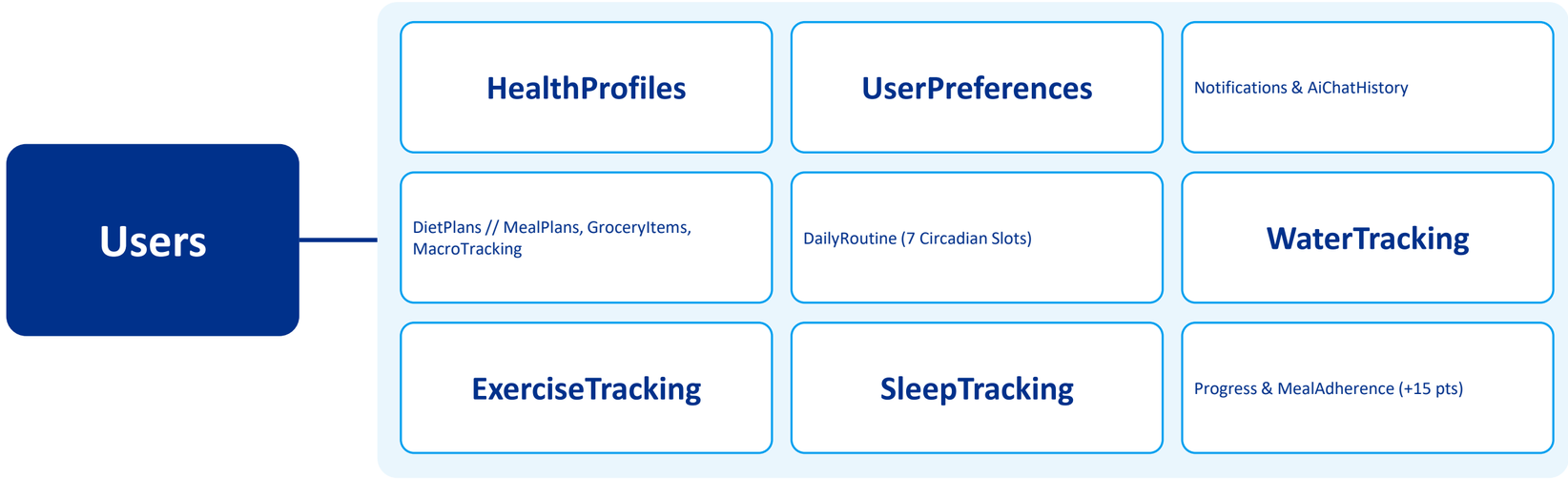
2 Select foods per meal

Filter by preference, remove allergens, rank by nutrient similarity, attach a recipe.

3 Produce outputs

Wellness score, grocery list, plan, recipes and substitutions.

Database Design



Food and recipe catalogue



Key relationships

- One user has one health profile.
- One user can have many diet plans.
- One diet plan contains many meal plans.
- Recipes contain multiple food ingredients.
- User-owned records are filtered using authenticated user ID.

Security, Privacy and Offline Support

Security plan

- Salted password hashes using bcrypt or Werkzeug
- JWT authentication for API requests
- HTTPS communication
- Server-side validation
- SQLAlchemy parameterised queries
- User-specific query filtering
- Secrets stored in environment variables

Planned backend stage

Offline support

- Store profile, meal plan, grocery list and logs locally
- Queue unsynchronised logs
- Synchronise when the network returns
- Require the server only for a new ML-generated plan

Local storage: prototype

Privacy

- Collect only required health information
- Do not present the output as a diagnosis
- Explain the limitations of guideline-derived recommendations

Design principle

Tools and Technologies

Layer	Technology	Purpose
Mobile	**Flutter and Dart**	Android mobile application
Backend	**Python and Flask**	REST API backend
Machine learning	**Scikit-learn**	Machine learning models
Data preparation	**Pandas and NumPy**	Dataset preparation
Database	**MySQL**	Relational database
Local storage	**SQLite or SharedPreferences**	Local mobile persistence
Security	**JWT and bcrypt**	Authentication and password security
Charts	**fl_chart**	Mobile analytics charts
Build	**GitHub Actions**	Cloud Android build
Testing	**Postman**	API testing
Development	**VS Code and Android Studio**	Development tools
Conversational AI	Google Gemini 1.5 Flash REST API	Intelligent wellness Q&A & dietary swaps
Local AI Engine	Heuristic Wellness Intel Engine	Zero-latency offline advisory system
PWA & Deployment	PWABuilder, WebAPK & Manifest v2	Android Play Store package (AAB) & iOS ready

Applications and Future Scope

Applications

- Personal diet and routine planning
- College and workplace wellness programmes
- Gym and fitness trainer support
- Sports student hydration and nutrition planning
- Preventive wellness support alongside professional medical advice

Future scope

- Health Connect and wearable integration
- Google Fit integration
- Barcode scanning
- Food recognition from images
- Regional Indian food datasets
- Multilingual support
- Dietitian review workflow
- Condition-aware plans under professional supervision
- Collaborative filtering after collecting enough real user data
- iOS support using the Flutter codebase

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Research papers

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Conclusion

What it is

This project develops a mobile wellness application that combines personalized diet planning, routine planning and daily habit tracking.

What it provides

BMR/BMI targets, 5-meal adaptive plans, Gemini AI coaching, 7-slot circadian routine schedules, macro balance monitoring, grocery categorization, habit steppers, wellness scoring, and admin governance.

How it is built

An offline-first PWA architecture with clean separation of presentation, domain, and persistence layers, packaged with WebAPK and service worker v9 for instant mobile and Play Store deployment.

How to use it

Designed as general wellness guidance, to be used together with professional medical or dietetic advice where required.

Thank You

Questions and discussion.

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